

How Much of LLM Trading-Agent Performance Survives Bias-Controlled Evaluation?

An Independent Pre-Registered Study with a Rationale Audit

Pre-Experiment Manuscript: Frozen Protocol and Analysis Plan

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This is a registered-report-style document: the experimental protocol is frozen and the analysis plan pre-committed before any experiment. All result fields are explicitly marked placeholders, and no empirical finding is claimed in this version.

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Abstract

Large language model (LLM) trading agents report backtests that outperform market benchmarks, while independent re-evaluations find these advantages disappear once evaluation biases are controlled — a contradiction the current literature cannot adjudicate. We present a pre-specified, bias-controlled evaluation of sentiment-driven LLM trading strategies on US large-cap equities. Six strategies — passive, classical-quant, classical-NLP, supervised-ML, LLM-scorer, and LLM-agent — are compared on identical data under rule-based universe selection, cost-adjusted next-open execution, multiple-testing correction, a three-condition memorization-quantification protocol, and an out-of-cutoff validation window. A complementary rationale audit tests whether the agent’s stated reasons are consistent with its trading behavior under controlled input perturbations. [RESULT SLOT 1: RQ1 headline — deflated-Sharpe outcome versus the passive benchmark.] [RESULT SLOT 2: memorization-gap estimate.] [RESULT SLOT 3: rationale-behavior consistency headline.] We release our code, cached model outputs, and frozen experimental protocol. (*Result sentences to be completed after experiments; this template asserts no findings.*)

Keywords: large language models; trading agents; backtest bias; memorization; evaluation; interpretability; financial NLP

1. Introduction

Large language model (LLM) trading agents have produced some of the most striking performance claims in recent computational finance. FinMem, an agent equipped with layered memory and an adjustable persona, reports leading trading performance relative to algorithmic baselines on real-world stock data (Yu et al., 2023). FinAgent, a multimodal successor, reports an average profit improvement of over 36% against twelve state-of-the-art baselines across six financial datasets (Zhang et al., 2024). TradingAgents, which organizes LLMs into the specialized roles of a simulated trading firm, likewise reports improvements in cumulative return, Sharpe ratio, and maximum drawdown over baseline models (Xiao et al., 2024). Yet when Li et al. (2025) re-evaluated LLM-based investing strategies within a unified backtesting framework spanning two decades and more than one hundred symbols, the reported advantages deteriorated substantially: the strategies proved overly conservative in bull markets, underperforming passive benchmarks, and overly aggressive in bear markets, incurring heavy losses. The literature therefore contains two directly contradictory claims about the same class of systems.

The contradiction is less surprising than it first appears, because the two bodies of evidence were produced under systematically different evaluation conditions. Quantitative finance has long documented that backtested performance is fragile: when many strategy configurations are trialed and only the best are reported, conventional performance statistics overstate true skill — a problem severe enough to motivate dedicated corrections such as the deflated Sharpe ratio (Bailey & López de Prado, 2014). LLM-based strategies inherit this fragility and add a technology-specific hazard. Although sentiment extracted by LLMs from news headlines has been shown to predict subsequent daily stock returns (Lopez-Lira & Tang, 2023), these models are trained on web corpora that include historical financial news; a backtest over a period contained in the training data is therefore contaminated both by look-ahead bias — the model may already know what followed the news — and by the distraction effect of entity-specific background knowledge (Glasserman & Lin, 2023).

An evaluation that controls neither multiple testing nor memorization cannot adjudicate between the optimistic and skeptical readings of the evidence.

No existing study, to our knowledge, combines the elements such an adjudication requires. Developer-reported evaluations rely on generally short backtesting periods, are largely confined to US and Chinese equities, and rarely account for trading costs (Ding et al., 2024); their reported advantages deteriorate when the cross-section of assets is broadened and the evaluation horizon extended (Li et al., 2025). The corrective re-evaluation of Li et al. (2025) broadens the cross-section and horizon but does not quantify memorization effects, does not include a supervised machine-learning baseline operating on the same information set, and does not test on data generated after the evaluated models’ training cutoffs. The present study is designed around all four missing elements: a protocol frozen before any experiment is run, an explicit memorization-quantification procedure adapted from anonymization-based debiasing (Glasserman & Lin, 2023), a graded ladder of baselines that isolates where any added value originates, and a secondary out-of-cutoff validation window.

A second gap concerns not how these agents perform but how they explain themselves. LLM agents produce natural-language rationales for their trades, and these rationales are commonly presented as an interpretability benefit of the paradigm (Yu et al., 2023). Yet research on chain-of-thought prompting demonstrates that model-generated explanations can systematically misrepresent the true basis of a model’s prediction (Turpin et al., 2023). Whether the stated rationales of trading agents are consistent with their trading behavior — and whether that consistency holds across market regimes — remains unexamined in the trading domain.

We therefore ask two questions. **RQ1:** How much of the reported outperformance of sentiment-driven LLM trading agents survives bias-controlled evaluation — rule-based universe selection, cost-adjusted next-open execution, memorization checks, and multiple-testing correction — on US large-cap equities? **RQ2:** Are the agents’ stated trading rationales consistent with their trading behavior, and how does this consistency vary across market regimes? Our working hypothesis is that reported outperformance is substantially attributable to evaluation artifacts and will not survive these controls. Both outcomes of RQ1 are informative: confirmation would establish that current evidence for LLM trading alpha is largely an evaluation artifact, whereas refutation would constitute the first bias-controlled demonstration of such alpha.

To answer these questions, we evaluate six strategies on an identical universe of large-capitalization US equities: an equal-weight buy-and-hold benchmark, a momentum rule, a FinBERT sentiment rule, a gradient-boosted-tree model trained on the same sentiment and technical features, an LLM sentiment scorer, and a full LLM agent that issues decisions with stated rationales. Evaluation follows a two-track design: a historical track with explicit memorization quantification, and a forward, out-of-cutoff validation track. Section 3 details the data and bias controls; Section 4 specifies the strategies and the pre-specified statistical protocol.

This paper makes four contributions. First, we provide a pre-specified, bias-controlled comparative evaluation of sentiment-driven LLM trading strategies against passive, classical-quant, classical-NLP, and supervised-ML baselines on identical data. Second, we quantify the contribution of training-data memorization to historical LLM backtest performance using entity-anonymization and date-shuffling controls. Third, we introduce a rationale audit — absent from prior work on trading agents — linking what these systems say to what they do across market regimes. Fourth, we release our code, cached model outputs, and evaluation harness to support reproduction and extension.

2. Related Work

This study sits at the intersection of four literatures: LLM trading agents, news sentiment and asset returns, backtest validity, and the faithfulness of model-generated explanations. We review each and identify the gap at their intersection.

2.1 LLM Trading Agents

Architectures in this area fall into three broad families. Memory- and reflection-driven single agents equip an LLM with structured storage and self-evaluation: FinMem combines a profiling module, layered memory, and a decision module aligned with the cognitive structure of human traders (Yu et al., 2023); FinAgent extends this design with multimodal inputs and dual-level reflection (Zhang et al., 2024). Multi-agent frameworks distribute reasoning across specialized roles: TradingAgents simulates a trading firm with fundamental, sentiment, and technical analysts, bull and bear researchers who debate market conditions, and a risk-management team (Xiao et al., 2024). A third family fine-tunes the model itself: Trading-R1 aligns LLM reasoning with trading principles through supervised fine-tuning and a staged reinforcement-learning curriculum, reporting improved risk-adjusted returns on six equities and exchange-traded funds (Xiao et al., 2025). All report favorable backtests. Across the field, however, the evaluation regime is thin: surveyed studies rely on generally short backtesting windows, are largely confined to US and Chinese equity markets, and rarely incorporate trading costs (Ding et al., 2024). When Li et al. (2025) re-evaluated LLM strategies over two decades and more than one hundred symbols within a unified framework, previously reported advantages deteriorated significantly, with strategies proving overly conservative in bull markets and overly aggressive in bear markets.

2.2 News Sentiment and Asset Returns

The proposition that textual sentiment carries return-relevant information predates LLMs by two decades. Tetlock (2007) showed that high media pessimism in a daily Wall Street Journal column predicts downward pressure on prices followed by reversion to fundamentals. Loughran and McDonald (2011) demonstrated that general-purpose sentiment dictionaries systematically misclassify financial text and constructed finance-specific word lists that became widely used for lexicon-based measurement. García (2013) found that the predictive content of news sentiment for daily returns is concentrated in recessions, establishing that sentiment effects are regime-dependent — a finding that directly motivates the regime analysis in the present study. Measurement technology has since advanced from lexicons to pretrained transformers, with FinBERT improving the state of the art on financial sentiment benchmarks (Araci, 2019), and most recently to general-purpose LLMs: Lopez-Lira and Tang (2023) showed that sentiment scores assigned by ChatGPT to news headlines predict subsequent daily stock returns. This progression supplies both the classical-NLP and LLM-scorer baselines in our strategy ladder, and its regime-dependence finding shapes our secondary research question.

2.3 Backtest Validity and LLM Memorization

A parallel literature documents why reported backtest performance so often fails to generalize. When many strategy configurations are evaluated and the best is reported, conventional performance statistics overstate skill; Bailey and López de Prado (2014) formalize the correction with the deflated Sharpe ratio (DSR), which adjusts for selection bias, backtest overfitting, and non-normality of returns. LLMs introduce an additional contamination channel absent from classical strategies. Glasserman and Lin (2023) distinguish two effects when a model’s training corpus overlaps the backtest period: look-ahead bias, in which the model may possess direct knowledge of returns that followed a news article, and a distraction effect, in which background knowledge about the company interferes with the sentiment-measurement task itself; using an anonymization procedure to separate the two, they find the distraction effect to be the larger. Building on this logic, Jeon and Lee (2026) propose an anonymization-first evaluation framework in which tickers and company names are masked before LLM agents make portfolio decisions, addressing memorization bias from pre-training alongside survivorship bias in backtest construction. Our memorization-quantification protocol adapts these procedures, and our statistical protocol applies the deflated Sharpe ratio that this literature prescribes but that developer-reported evaluations have not adopted.

2.4 Faithfulness of Model-Generated Explanations

Trading agents present natural-language rationales as an interpretability benefit of the paradigm (Yu et al., 2023). Whether such rationales can be trusted is a separate empirical question. Turpin et al. (2023) demonstrate that chain-of-thought explanations can systematically misrepresent the true basis of a model’s prediction: when biasing features are introduced into inputs, models change their answers while producing plausible explanations that never mention the manipulation. The stated rationale of a trading agent — for example, that a position was taken on the basis of news sentiment — therefore cannot be assumed to reflect the actual driver of the decision. To our knowledge, no prior work has tested rationale–behavior consistency for trading agents by perturbing their inputs and observing whether decisions change in the manner the rationales imply; this is the audit our second research question supplies.

2.5 Synthesis

Classical finance theory frames what is at stake. Under the efficient markets hypothesis, publicly available news should be impounded into prices too quickly for a daily sentiment strategy to profit net of costs (Fama, 1970); the adaptive markets hypothesis instead predicts that the profitability of any such strategy will vary with market conditions as competition and arbitrage evolve (Lo, 2004) — consistent with the regime-dependence documented by García (2013) and the bull/bear asymmetry found by Li et al. (2025). Each literature reviewed above supplies one ingredient of a credible adjudication: candidate systems (§2.1), validated signal sources and baselines (§2.2), bias controls and corrected statistics (§2.3), and constructs for auditing explanations (§2.4). No existing study combines them: developer evaluations lack the controls, the corrective re-evaluation of Li et al. (2025) applies neither memorization quantification nor a supervised-ML baseline on an identical information set, and the faithfulness literature has not reached trading. The present study occupies this intersection.

3. Methodology: Data and Bias Controls

Every design choice in this section is motivated by a named threat to validity. We describe the data and universe (§3.1), the temporal-integrity procedures addressing look-ahead bias and leakage (§3.2), the point-in-time procedures addressing survivorship bias (§3.3), the protocol that measures training-data contamination (§3.4), and the validation steps — including a pilot study — that precede full-scale experimentation (§3.5). Dataset properties that can only be established empirically are marked as verification items and confirmed during the pilot rather than assumed. Figure 1 summarizes the end-to-end pipeline that implements these controls.

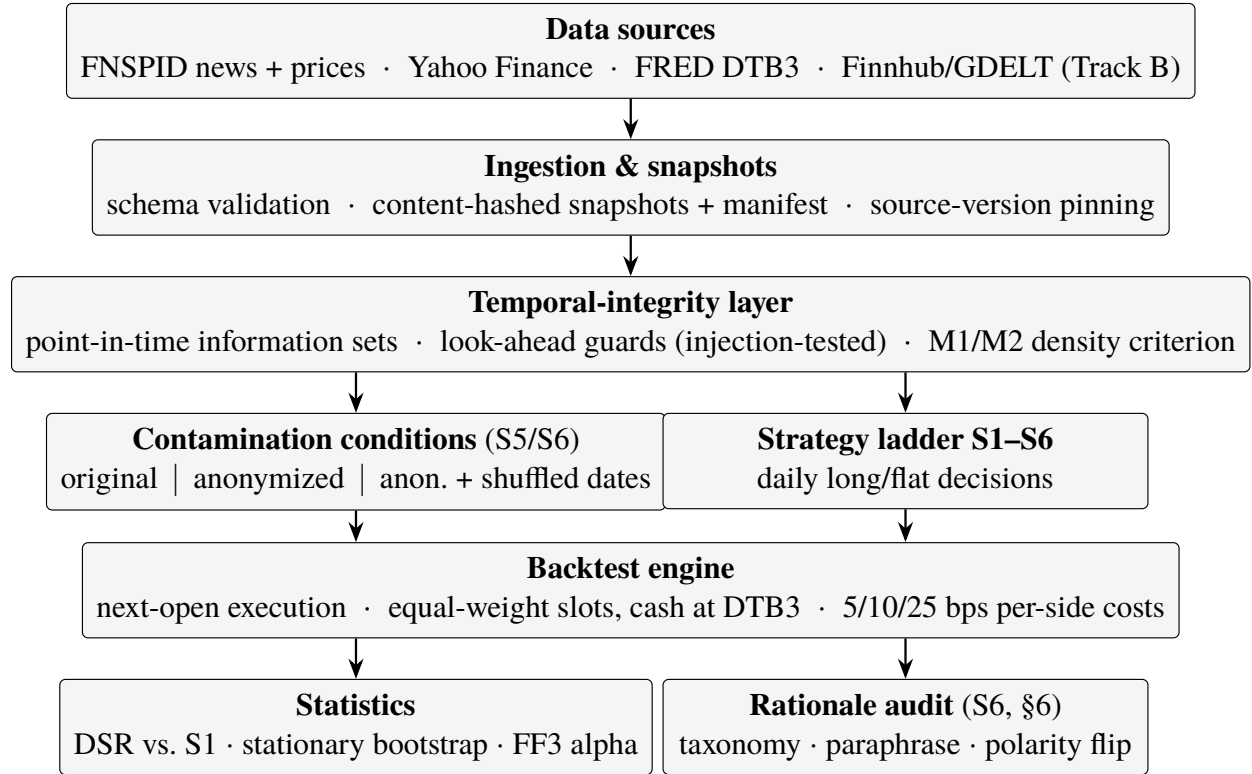


Figure 1: End-to-end architecture of the evaluation pipeline. Data flows from source acquisition through content-hashed snapshotting and the injection-tested temporal-integrity layer; the LLM-based strategies (S5/S6) additionally pass through the three contamination conditions of §3.4 before all strategies meet the identical backtest engine and pre-specified statistics.

3.1 Data Sources and Universe

Track A (historical, primary). Our primary news and price source is the Financial News and Stock Price Integration Dataset (FNSPID), which combines 29.7 million stock-price records with 15.7 million time-aligned news records for 4,775 S&P 500 companies over 1999–2023, drawn from four financial news websites (Dong et al., 2024). We use FNSPID because it is, to our knowledge, the largest freely available corpus in which news and prices are already time-aligned, which reduces the alignment errors that arise when researchers join news and price data themselves. It carries

three risks for our design. First, its news is drawn from four websites and therefore samples the news environment rather than exhausting it; we treat our results as conditional on this information set and report per-ticker article counts (Table 1, to be populated after the pilot) so readers can assess coverage. Second, the density and timestamp granularity of its news coverage for our universe and window are unmeasured dataset properties; both are pilot verification items governed by the pre-specified adequacy criterion of §3.5, and the timestamp question is consequential enough that the execution rule of §3.2 is designed to remain conservative under the worst case. Third, the dataset ends in 2023, inside the training window of any current LLM; this is not a disqualification but the central measurement problem of the study, addressed in §3.4. The Track A evaluation window is January 2022 through December 2023, chosen to include both a bear market (2022) and a recovery (2023) while remaining recent enough for adequate news density [verification item, §3.5 criterion].

<i>To be populated after experiments.</i>

Table 1: Universe and news-coverage statistics: per-ticker article counts and M1/M2 density metrics (placeholder; to be populated after the pilot).

Track B (forward, secondary). To obtain data that post-dates the pinned model’s training cut-off, we assemble a six-month window (January–June 2026) with daily prices from Yahoo Finance and news from two free sources: the Finnhub company-news API (Finnhub, n.d.) and the GDELT project, which monitors global news media and provides timestamped, machine-readable records (Leetaru & Schrod, 2013). Because GDELT is designed for event and tone monitoring rather than firm-level news retrieval, its precision when filtered to specific companies is a verification item distinct from raw coverage. Track B exists because no anonymization procedure can fully guarantee that memorized knowledge has been neutralized; genuinely out-of-cutoff data is the only evaluation for which memorization is impossible by construction. Its cost is scale: the window is short, and free news APIs’ coverage for our universe is a verification item. Track B is therefore scoped as a validation check on Track A conclusions, not as an independent test with its own statistical power. Figure 2 lays out the three study windows on the calendar.

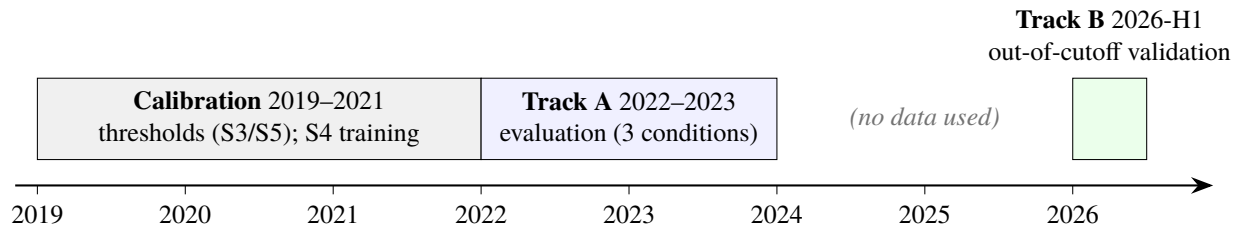


Figure 2: Study data windows. Calibration (2019–2021) supplies thresholds and S4 training data; Track A (2022–2023) is the primary evaluation window, run under the three contamination conditions of §3.4; Track B (2026-H1) post-dates the pinned model’s training cutoff, so memorization is precluded by construction. Data from 2024–2025 are not used.

Universe. The universe is the twenty largest S&P 500 constituents by market capitalization as

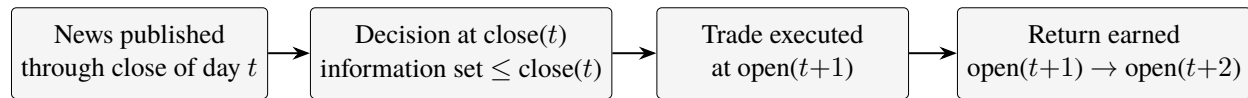
of December 31, 2021 — the last trading day before the Track A window opens — selected by rule and frozen before any experiment. Rule-based selection at the window’s start reduces the risk of selection bias, because the rule uses no information from the evaluation period. Mega-capitalization stocks are a deliberately conservative testbed: they are the most heavily covered and most liquid corner of the market, and the sentiment literature finds predictive content concentrated in stressed conditions rather than in such stocks generally (García, 2013). A null result here therefore does not preclude sentiment alpha elsewhere; Section 8 returns to this point. Table 2 consolidates the data sources, their roles, and the study windows.

Table 2: Data sources, roles, and study windows. Calibration window 2019-01-02 to 2021-12-31; Track A evaluation window 2022-01-03 to 2023-12-29; Track B validation window 2026-01-02 to 2026-06-30.

Source	Role	Windows served
FNSPID (news + prices)	Track A news corpus, time alignment	calibration, Track A
Yahoo Finance (yfinance)	Daily prices; S&P 500 TR benchmark	all
FRED series DTB3	Risk-free rate (3-month T-bill)	all
Cboe VIX	Volatility-regime terciles	Track A
Kenneth French data library	Fama–French factor returns	Track A
Finnhub + GDELT	Out-of-cutoff news	Track B

3.2 Temporal Integrity

The information set for any decision executed at the open of trading day $t+1$ is restricted to data timestamped on or before the close of day t . Decisions are executed at the next session’s opening price, never at the same session’s close, because a close-of-day execution would implicitly assume the strategy could trade on news published up to the instant of the close — the classic look-ahead error in daily-frequency backtests. Figure 3 depicts this information-flow contract.



*Any record timestamped after close(t) cannot reach the decision:
the harness raises an error rather than filtering silently (validated by injection tests).*

Figure 3: The information-flow contract that removes look-ahead by construction. A decision may use only records timestamped at or before the close of day t ; it is executed at the next open and earns the following open-to-open return.

This procedure reduces the risk of look-ahead bias; it cannot eliminate it entirely, because it depends on the accuracy of source timestamps. Two safeguards address that dependency. First, the pipeline enforces the information-set rule mechanically: the backtest harness raises an error if any input record’s timestamp postdates the decision time, rather than relying on correct upstream filtering. Second, if the pilot finds that FNSPID news timestamps are date-only rather than intraday [verification item], we adopt the conservative convention that an article dated day t may have

appeared as late as that day’s close and is therefore first usable for the day $t+1$ open decision — sacrificing signal freshness to protect integrity; under intraday granularity, the exchange close is applied as the same-day cutoff instead (Appendix D). For the supervised baseline (S4, §4.1), all model selection and hyperparameter tuning use only data preceding the evaluation window, so that no evaluation-period information leaks into training.

3.3 Survivorship and Point-in-Time Correctness

Survivorship bias is a documented source of spurious performance in LLM trading backtests (Jeon & Lee, 2026). It enters daily backtests through two doors: selecting the universe from firms known to have survived the evaluation period, and using price series that silently drop delisted firms. Our universe rule closes the first door by construction, since selection occurs before the window using only prior information. For the second, the practical exposure is limited — the twenty largest S&P 500 firms as of end-2021 — but not zero, and we handle it explicitly: if any universe member is delisted or acquired during the window, its position is liquidated at the last available price, the proceeds are held at the risk-free rate, and the event is reported. Whether FNSPID and Yahoo Finance retain full histories for any such firm, and whether adjusted prices correctly reflect splits and dividends across our window, are verification items checked in the pilot against a second, independent price source.

3.4 Memorization and Contamination Controls

Because the Track A window lies inside the training corpora of current LLMs, an LLM-based strategy’s historical performance conflates genuine signal extraction with recall of memorized outcomes. Glasserman and Lin (2023) established the diagnosis and the core instrument — anonymizing company identifiers separates what a model infers from the text from what it retrieves about the named firm — and Jeon and Lee (2026) extended anonymization from sentiment scoring to agentic portfolio decisions. Our protocol inherits both elements and differs in purpose and scope in three respects. First, where prior work deploys anonymization primarily as mitigation, we deploy it as measurement: each LLM-based strategy (S5 and S6, §4.1) is run under three pre-specified conditions — original inputs, anonymized inputs, and anonymized inputs with shuffled dates — and the performance differences across conditions are themselves reported quantities, estimating how much of apparent performance memorization explains. Second, the protocol is applied uniformly across a graded strategy ladder rather than to a single system, allowing us to ask whether memorization effects grow with the model’s degrees of freedom. Third, the historical estimate is cross-checked against Track B, where memorization is precluded by construction; agreement between the tracks would support the protocol’s adequacy, and disagreement would itself be an informative finding. We claim for this design a difference in emphasis and integration, not a wholly new instrument. Figure 4 maps each named threat to validity onto its pre-specified control.

3.5 Validation Procedures and Pilot Study

Full-scale evaluation is preceded by a staged validation. First, the backtest harness is unit-tested: the portfolio accounting identity (§4.3) is verified on synthetic price paths with known outcomes, and the temporal-integrity assertion of §3.2 is tested by deliberately injecting future-dated records

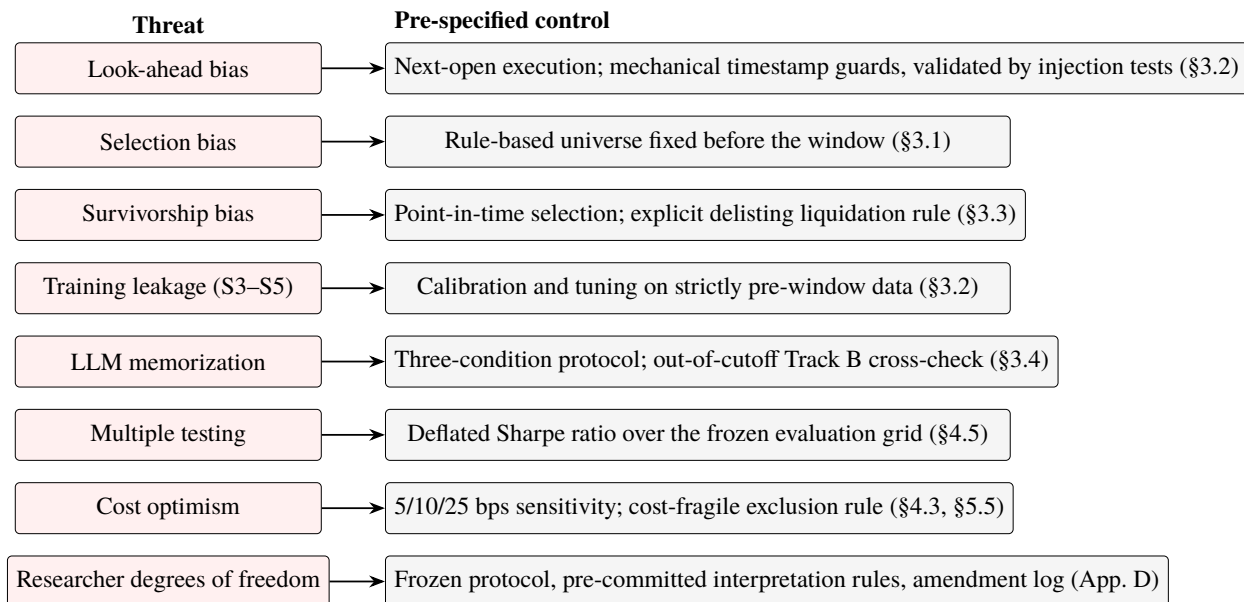


Figure 4: The bias-control framework. Each threat to validity is paired with the pre-specified control that addresses it and the section defining that control.

and confirming the harness rejects them. Second, the passive and momentum baselines (S1, S2) are run over the full window and reconciled against an independent implementation of the same rules; disagreement beyond rounding indicates an accounting error, not a finding. Third, a pilot study — two tickers over a three-month sub-window — exercises the complete pipeline, including LLM calls, caching, and the anonymization conditions. The pilot evaluates pipeline integrity, data availability, and implementation correctness before large-scale evaluation; it is not an evaluation of strategy performance, and its trading outcomes will not be reported as evidence for or against any hypothesis. Its verification targets are: FNSPID news density for the universe and window (under the criterion below); FNSPID timestamp granularity; delisted-firm and corporate-action handling in both price sources; Finnhub and GDELT coverage for Track B; and per-decision LLM cost and latency, which set the feasibility envelope of the full run.

News-density adequacy criterion (frozen before the pilot). Adequacy is assessed by two metrics computable directly from the data. *M1 (availability)*: the fraction of trading days on which at least one timestamp-valid news observation for the ticker exists within the trailing three trading days — the same news window the text-based strategies consume — available before that day’s close. *M2 (timestamp validity)*: the fraction of a ticker’s articles carrying a usable timestamp of at least daily granularity. Decision rule: a ticker passes if $M1 \geq 0.90$ and $M2 \geq 0.95$; failing tickers are flagged as insufficiently covered and excluded from the primary analysis universe, with all exclusions reported; if more than five of the twenty universe tickers fail, Track A’s news source is deemed inadequate, triggering a documented protocol amendment (source augmentation or halt) before any strategy is run. Rationale: the M1 bound permits occasional quiet periods while ensuring that the information-flow assumption underlying daily text-based strategies holds on nearly all decision days, and its trailing-window form ties the criterion to what the strategies actually read rather than to raw article counts, which vary with firm popularity; the M2 bound makes the temporal-integrity rule of §3.2 enforceable. Both thresholds are frozen here, before the pilot, and are never

revised in light of pilot or experimental outcomes. The criterion measures availability of timestamp-valid information only; it deliberately does not measure the informativeness of the news, so that low signal quality is treated as a potential finding, never as grounds for excluding data.

Pilot findings that alter the design are documented as protocol amendments before the full experiment begins (Appendix D). For reproducibility, we record exact version identifiers of all data sources — the FNSPID repository commit and the query dates and parameters for Yahoo Finance, Finnhub, and GDELT — and release the processing code and cached LLM outputs, so the study’s inputs can be reconstructed even where third-party data cannot be redistributed directly.

4. Experimental Design: Strategies and Evaluation Protocol

Where Section 3 established that the data can be trusted, this section establishes that the comparison is fair. All strategies operate under identical conditions — the same universe, window, information set, execution rules, cost assumptions, and portfolio constraints — so that observed performance differences are attributable to the decision rule and to nothing else. All statements in this section describe procedures to be executed; no empirical outcome is asserted.

4.1 Strategy Definitions

The six strategies form a graded ladder in which each adjacent pair differs, as far as practicable, in a single ingredient. This structure is the study’s central inferential device: rather than asking only whether the most complex system outperforms the simplest benchmark, it asks *where along the ladder* any added value enters — with the price signal, the text signal, the learning algorithm, the language model, or the agent architecture. Strategies S1–S4 are baselines, S5 is an ablation variant, and S6 is the system under test. No ordering of expected performance is implied. Figure 5 depicts the ladder and the single ingredient each rung adds.

S1 — Equal-weight buy-and-hold (passive benchmark). All twenty universe stocks are bought in equal weights at the start of the window and held. S1 answers the question every active strategy must face: does the decision rule add anything over doing nothing?

S2 — Time-series momentum (classical price baseline). A stock is held long when its trailing twelve-month return, excluding the most recent month, is positive, and moved to cash otherwise, with signals refreshed monthly. The specification follows the momentum evidence of Jegadeesh and Titman (1993), including its conventional monthly signal frequency; the resulting difference in rebalancing frequency relative to the daily strategies is a property of the strategy itself, is charged costs under the identical per-trade model, and is made visible through the turnover metric (§4.3). S2 asks whether a decades-old price-only anomaly explains any advantage attributed to more elaborate systems.

S3 — FinBERT sentiment rule (classical NLP baseline). Each trading day, all universe-relevant articles are scored by FinBERT (Araci, 2019) and averaged per ticker over the trailing news window; a stock is held long the following session if this sentiment exceeds a threshold calibrated on pre-window data only. The threshold is the Sharpe-maximizing percentile of the pooled pre-window score distribution from the frozen grid {30, 40, 50, 60, 70}, with ties resolved to the lowest percentile; ticker-days without news map to flat (Appendix D). S3 asks whether pre-LLM sentiment technology suffices to capture the text signal.

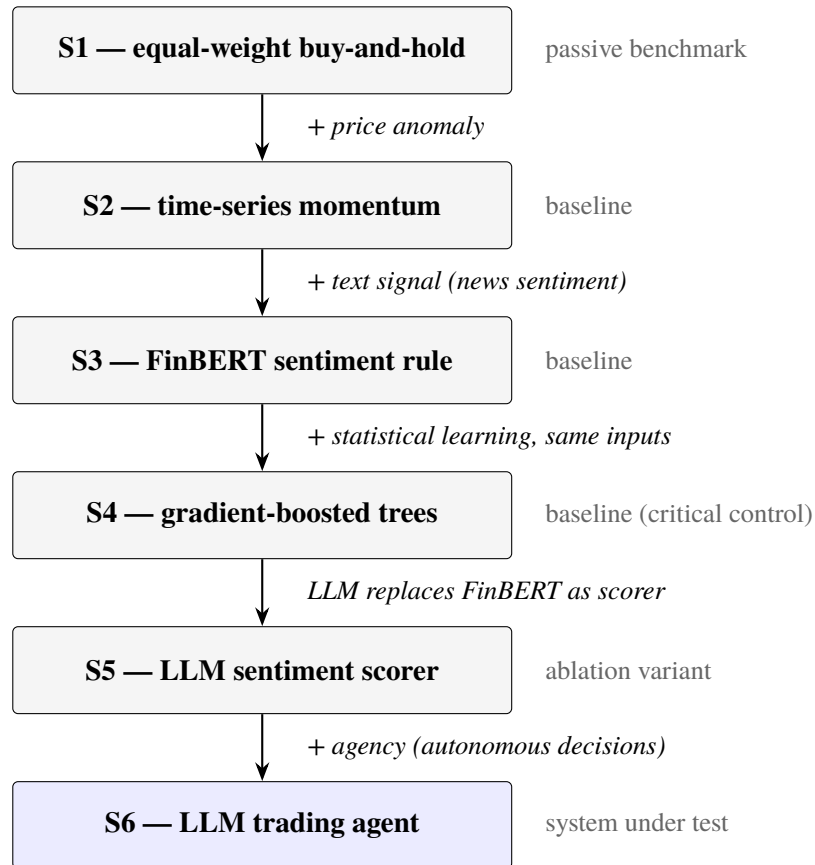


Figure 5: The graded strategy ladder. S1–S4 are baselines, S5 is an ablation variant, and S6 is the system under test; each adjacent pair differs, as far as practicable, in a single ingredient.

S4 — Gradient-boosted trees (supervised-ML baseline). A gradient-boosted tree classifier is trained — on data strictly preceding the evaluation window — to predict the direction of the open-to-open return its decision would earn (open of $t+1$ to open of $t+2$, matching the execution contract of §4.3) from the same FinBERT sentiment features plus standard technical features (lagged returns, moving-average ratios, realized volatility, volume). Its predictions map to the same long/flat action space via the frozen rule: long iff $P(\text{up}) > 0.5$. S4 is the ladder’s critical control: without it, any advantage of S5 or S6 over S3 could not be attributed to the language model rather than to statistical learning in general.

S5 — LLM sentiment scorer (ablation variant). The pinned open-weight LLM (§4.7) assigns each ticker-day a sentiment score from the same news inputs, and the score feeds a threshold rule calibrated by the identical procedure as S3 — on pre-window data only — since the two scorers’ output scales differ and sharing a numeric threshold would not yield a fair comparison. The S3–S5 pair isolates measurement technology (transformer classifier versus generative LLM); the S5–S6 pair isolates agency (scoring versus autonomous decision-making).

S6 — LLM reference agent (system under test). Each trading day, per ticker, the same LLM receives a prompt containing recent news summaries, a price-history window, and the current position, and outputs a long/flat decision together with a natural-language rationale, which is logged for the audit of Section 6. S6 is a deliberately simplified single-agent design in the style of FinMem (Yu et al., 2023): it omits fine-tuning, multimodal inputs, and multi-agent debate. Conclusions therefore attach to this reference class, not to any named published system; published systems’ reported results are discussed as context in §5.11.

4.2 Experimental Controls

The following are held constant across all six strategies: the universe and its selection rule (§3.1); the evaluation window; the news corpus and the information-set rule (§3.2); the long/flat action space with no shorting and no leverage; equal-weight position sizing; execution at the next session’s open; the transaction-cost model (§4.3); the treatment of uninvested capital, which earns the risk-free rate, proxied by the three-month US Treasury bill rate (FRED series DTB3; Federal Reserve Bank of St. Louis, n.d.); and the contamination conditions of §3.4, which apply to the LLM-based strategies. Strategies differ only in the mapping from the shared information set to the daily action vector. Table 3 consolidates the frozen experimental settings.

Table 3: Frozen experimental settings (Appendix D). No value changes after results are seen except through the Amendment Log.

Setting	Frozen value
Universe	20 largest S&P 500 constituents by market cap, 2021-12-31
Action space	long/flat; no leverage, no shorting; equal-weight slots
Execution	decisions at close of t ; fills at open of $t+1$
Transaction costs	5 / 10 / 25 bps per side (10 bps headline)
Contamination conditions	original; anonymized; anonymized + shuffled dates
Model instrument	Qwen3-14B, bf16 (vLLM); $T=0.7$, top- p 0.8 (A-001); seeds 11/42/2026
Bootstrap	stationary; $B=10,000$; expected block 21 days (5/63 sensitivity)
Inference budget	USD 200 total; 30% reserved for the rationale audit

4.3 Portfolio Accounting, Execution, and Transaction Costs

Daily portfolio return is defined by a single accounting identity applied to every strategy: the equal-weighted mean of open-to-open returns over stocks held long, plus the risk-free return on the cash weight, minus transaction costs incurred by that day’s position changes. Strategies issue their entry decisions on the last trading day before the evaluation window, so every strategy is invested (or deliberately flat) from the window’s first open — the like-for-like entry that S1’s definition requires (Appendix D). Turnover is the sum of absolute weight changes at each rebalancing, reported annualized. Costs are modeled as a proportional charge of 10 basis points (bps) per side, covering commissions, half-spread, and slippage jointly; because any single number is contestable, all headline analyses are repeated at 5 and 25 bps and reported in full. Market impact is assumed negligible — defensible only because the universe consists of the most liquid US equities and the notional portfolio is assumed small relative to their daily volume; this assumption is restated as a limitation in Section 8. No commission-free execution is assumed to justify a zero-cost setting, since spread and slippage exist regardless of commission structure.

4.4 Evaluation Metrics

We report, for each strategy and condition: **cumulative return** and **annualized return**, the quantities most prior work emphasizes, included for comparability; **annualized volatility**, without which return figures are uninterpretable; the **Sharpe ratio** (Sharpe, 1966), a risk-adjusted performance measure subject to the limitations that it treats upside and downside variability symmetrically and overstates skill when many configurations are trialed — the latter being the reason it is not our primary endpoint; **maximum drawdown** and **downside deviation** — the annualized square root of the mean squared negative daily return, the standard Sortino denominator — which capture the asymmetric loss experience that volatility alone misses and which prior work identifies as a specific weakness of LLM strategies in bear markets (Li et al., 2025); and a **factor-adjusted alpha** from a Fama–French three-factor regression (Fama & French, 1993), using factor data from the Kenneth French data library (French, n.d.), because a strategy can appear to add value while merely loading on known risk factors. Table 4 summarizes the metric suite and the role each metric plays.

Table 4: Evaluation metrics and their pre-specified roles (§4.4–§4.5).

Metric	Role
Cumulative / annualized return	comparability with prior work
Annualized volatility	scale for interpreting returns
Sharpe ratio	risk-adjusted performance (not primary)
Deflated Sharpe ratio vs. S1	primary endpoint ; multiple-testing corrected
Maximum drawdown; downside deviation	asymmetric loss experience
Fama–French three-factor alpha	separates value-add from factor loadings
Annualized turnover	exposes cost burden and rebalancing intensity

Benchmark comparisons are made against S1 and against the S&P 500 total-return index (Appendix D records the data source); outperformance will be assessed relative to these benchmarks, not asserted in advance. S1 answers whether a strategy beats passive holding of the identical

universe under identical assumptions — the strictest like-for-like comparison — while the index comparison contextualizes results against the broad-market alternative an investor actually faces; using both avoids the unfair advantage of comparing a twenty-stock strategy only against a benchmark it does not track.

4.5 Statistical Protocol and Precision Statement

The primary endpoint is the **deflated Sharpe ratio (DSR)** of each active strategy relative to S1 (Bailey & López de Prado, 2014), computed with the number of trials set to the count of all strategy–cost–condition combinations actually executed under the pre-specified grid (six strategies at three cost settings, with the three contamination conditions of §3.4 applied to the LLM-based strategies), so that the correction reflects everything run, not only what is highlighted. Uncertainty for Sharpe-ratio differences is quantified with stationary-bootstrap confidence intervals (Politis & Romano, 1994), chosen over the i.i.d. bootstrap because daily returns exhibit serial dependence that resampling must preserve. All runs in the grid are reported; no ticker, window, or configuration is dropped after results are seen.

We state the study’s precision limits in advance. With twenty assets over a two-year primary window, statistical power against modest performance differences is limited: confidence intervals are expected to be wide, and the absence of a statistically significant difference will be interpreted as the absence of evidence, not as evidence of equivalence. All point estimates are reported with uncertainty, and no claim of outperformance or underperformance will be made from point estimates alone.

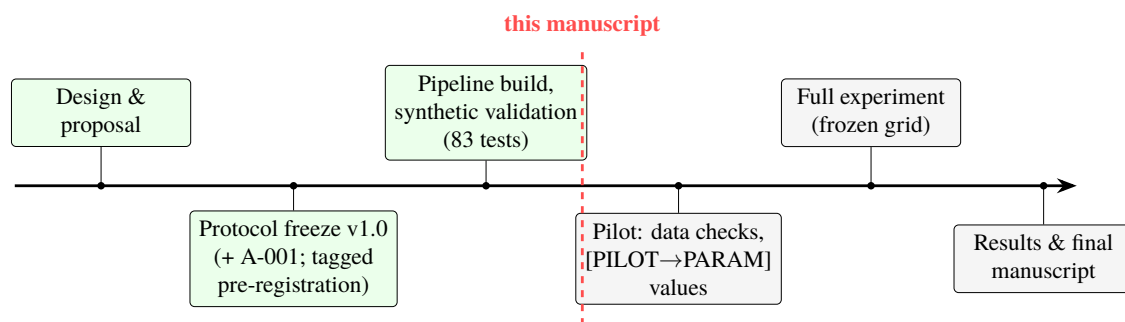
4.6 Regime Analysis

Because the sentiment literature finds predictive content concentrated in particular market conditions (García, 2013), and the adaptive markets hypothesis predicts time-varying strategy profitability (Lo, 2004), performance is additionally reported within market regimes. Regimes are defined by rules fixed before any experiment: (i) *bear* versus *non-bear* months, where a month is bear if the S&P 500 closes that month more than 20% below its prior running peak, and (ii) volatility terciles of the evaluation window based on the Cboe VIX index level (Cboe Global Markets, n.d.), with the in-window construction of terciles acknowledged as a descriptive convenience. The regime analysis is exploratory and descriptive: per-regime observation counts are too small to support hypothesis testing, no per-regime significance claims will be made, and regime definitions will not be revised after results are seen. Its role is to characterize *when* strategies succeed or fail, generating hypotheses for confirmatory testing in future work.

4.7 Reproducibility

All LLM-based strategies use a single pinned open-weight model, run locally, with the exact model identifier, weights revision, quantization, and inference parameters recorded in the frozen protocol; the model is selected and fixed before experiments begin. Decoding follows the vendor-documented inference configuration — temperature 0.7 and top-p 0.8, adopted under protocol amendment A-001 (Appendix D) because the model’s documentation advises against greedy decoding — with reproducibility enforced instead through three fixed seeds per configuration (across-seed dispersion

reported), full call caching, and revision pinning. Structured outputs are validated against JSON schemas; a malformed response is retried once and then logged as an abstention (the position is held and the abstention counted). Every model call — prompt, parameters, and output — is cached and released, allowing exact reconstruction of all LLM decisions without re-running inference. Experiments are driven by version-controlled configuration files; each run logs its configuration hash, data version identifiers (§3.5), and outputs to an append-only record. The full protocol — prompts, thresholds, hyperparameter search spaces for S4, and the evaluation grid — is frozen and archived (Appendix D) before the first full-scale experiment, and any post-freeze amendment is documented with its rationale in the Amendment Log. Figure 6 places the frozen protocol within the study timeline.



Shaded milestones are complete; unshaded milestones are pending. No experiment precedes a fully valued protocol lock (App. D).

Figure 6: Study timeline. The protocol was frozen — and pre-registered via a tagged repository commit — before any experiment; this manuscript stands at the dashed marker, ahead of the pilot and all empirical runs.

5. Results: Pre-Specified Reporting and Analysis Framework

Registered-report note: experiments have not yet been conducted. This section specifies, in advance, what will be measured, how it will be calculated, how outputs will be reported, and which interpretations the design does and does not license. Every table and figure is a placeholder to be populated exclusively from executed runs. Nothing here states, estimates, or predicts an empirical outcome. All pre-committed analytic choices are archived in Appendix D before the first full-scale run, so that no choice can be adjusted, consciously or otherwise, after results are seen.

5.1 Primary Strategy Comparison

<i>To be populated after experiments.</i>

Table 5: Strategy performance comparison, S1–S6, original-input condition at 10 bps (placeholder; to be populated after experiments).

The pre-specified content of Table 5 — Rows: the six strategies under the original-input condition at the 10-bps cost setting. Columns: cumulative return, annualized return, annualized volatility, Sharpe ratio, DSR versus S1, maximum drawdown, downside deviation, annualized turnover, and Fama–French three-factor alpha with its standard error. Every cell derives from the accounting identity of §4.3 applied to logged daily positions; no quantity is computed outside the harness.

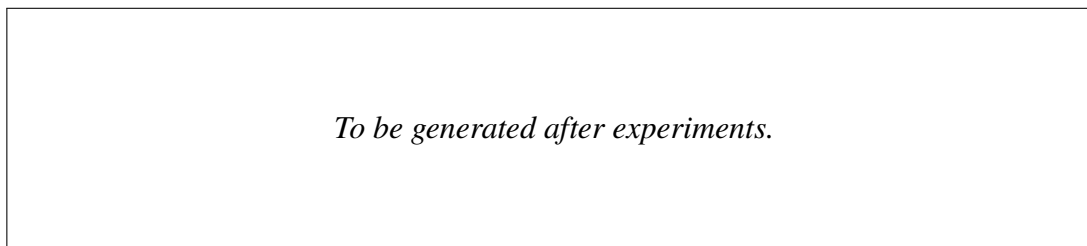


Figure 7: Equity curves for S1–S6 and the S&P 500 total-return index over the Track A window, log scale, regime shading per §4.6 (placeholder; to be generated after experiments).

Figure 7 will present these equity curves, plotted on a log scale over the Track A window with regime shading as defined in §4.6.

The first sentence of the populated section will state the answer to RQ1 in terms of the primary endpoint: whether any active strategy’s DSR versus S1 is positive and statistically distinguishable from zero after the multiple-testing correction of §4.5. The evaluation will test whether the LLM-based strategies provide improved risk-adjusted performance relative to the baselines under the defined experimental conditions; the design treats improvement, equivalence, and deterioration as equally reportable outcomes.

5.2 Benchmark and Risk-Adjusted Comparison

Each active strategy will be compared against both benchmarks of §4.4 under pre-committed interpretation rules. A claim that a strategy “adds value” requires all three of: a positive DSR versus S1 after correction; a three-factor alpha distinguishable from zero; and survival of the result at the 25-bps cost setting. Failing any one, the result will be described in the specific terms of what was and was not observed — for example, a positive raw Sharpe difference with a DSR indistinguishable from zero — without a value-added claim. Point estimates will never be interpreted without their interval estimates.

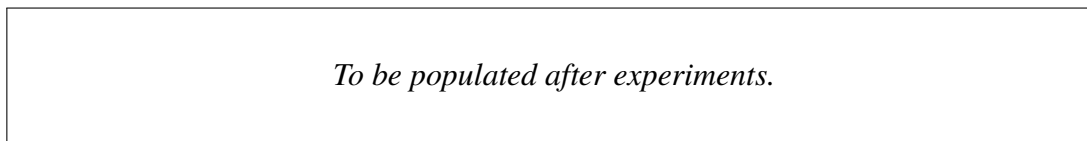


Table 6: Stationary-bootstrap 95% confidence intervals for Sharpe-ratio differences versus S1 (placeholder; to be populated after experiments).

Table 6 — Stationary-bootstrap intervals (§4.5) at the 95% level, with the block-length parameter recorded in Appendix D.

5.3 Ablation Analysis

The ladder contrasts of §4.1 will be reported as pairwise differences with intervals: S2—S1 (price anomaly), S3—S1 (classical text signal), S5—S3 (measurement technology: generative LLM versus FinBERT), S4 versus S5 (supervised learning versus LLM scoring on the same information set), and S6—S5 (agency versus scoring). Each contrast answers one attribution question, and conclusions will name the rung, not the paradigm: a positive S6—S5 contrast would support a contribution of the agent architecture *given this model and window*, not a general claim that “agents work.” Contrasts whose intervals include zero will be reported as uninformative on that rung.

5.4 Memorization and Contamination Results

<i>To be populated after experiments.</i>

Table 7: LLM-based strategies under the three contamination conditions of §3.4, with the memorization-gap estimand (placeholder; to be populated after experiments).

Table 7 — For S5 and S6: all Table 5 metrics under each condition, plus the pre-specified memorization-gap estimand — the difference in Sharpe ratio between original and anonymized conditions, with bootstrap intervals. Interpretation rules: a performance advantage present under original inputs but absent under anonymization will be attributed to memorization or entity-knowledge effects rather than to signal extraction, following Glasserman and Lin (2023); in the shuffled-dates condition, where no genuine time-aligned signal should exist, a Sharpe-ratio excess over S1 whose bootstrap interval excludes zero would indicate residual leakage and trigger a documented investigation before any Track A conclusion is drawn. This table stands as a contribution regardless of the direction of the RQ1 result.

5.5 Transaction-Cost Sensitivity

<i>To be generated after experiments.</i>

Figure 8: Sharpe ratio and cumulative return of each strategy at 5, 10, and 25 bps per side (placeholder; to be generated after experiments).

Figure 8 will report each strategy at every cost setting. The populated text will report which, if any, qualitative conclusions of §5.1–§5.3 change across cost settings. A conclusion that holds only at

the most favorable cost setting will be labeled cost-fragile and excluded from the paper’s headline claims.

5.6 Out-of-Cutoff Validation (Track B)

Track B results will be reported in the Table 5 format, clearly labeled as a short-window validation with limited power (§3.1). The pre-committed use is directional consistency, not significance: agreement in sign and approximate magnitude between Track A (anonymized condition) and Track B will be reported as supporting the adequacy of the anonymization protocol; disagreement will be reported as a finding about the protocol itself, and Track A conclusions will be correspondingly qualified. Track B will not be used to rescue or overturn Track A conclusions on its own.

5.7 Regime-Based Reporting

<i>To be populated after experiments.</i>
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Table 8: Per-regime descriptive performance with observation counts (placeholder; to be populated after experiments).

Table 8 — All strategies’ returns, volatility, drawdown, and — for S6 — position frequencies, within the bear/non-bear and VIX-tercile partitions of §4.6, with per-regime observation counts stated in the table. Consistent with §4.5, this reporting is descriptive: no per-regime significance tests will be conducted, no per-regime claims will appear in the abstract or conclusions, and any regime pattern will be framed as hypothesis-generating. The populated text will note whether observed patterns are directionally consistent with the regime-dependence documented by García (2013) and the bull/bear asymmetry reported by Li et al. (2025), without asserting confirmation of either.

5.8 Robustness Checks

Four pre-specified checks will be reported: across-seed dispersion for the LLM-based strategies (§4.7), with any configuration whose across-seed Sharpe range exceeds its bootstrap interval width flagged as unstable; per-ticker results for all strategies (Appendix C), reported in full to preclude selective emphasis; threshold-sensitivity analysis for S3 and S5, re-running each with thresholds at the 30th–70th percentiles of the pre-window score distribution and reporting the full performance curve — a grid fixed before experiments and chosen over a multiplicative perturbation of the calibrated value because percentile placement is invariant to each scorer’s arbitrary scale and remains meaningful when the calibrated threshold lies near zero, where a proportional perturbation would be negligible; and pilot-to-full consistency, confirming that no protocol amendment recorded under §3.5 altered any pre-specified analysis.

5.9 Statistical Framework and Limits of Inference

All interval estimates use the stationary bootstrap (Politis & Romano, 1994) on daily returns. The multiple-testing correction enters through the DSR (Bailey & López de Prado, 2014), applied as specified in §4.5 with the trial count fixed by the frozen grid; the trial count and all DSR inputs (skewness, kurtosis, sample length) will be reported so the calculation can be reproduced. Two limits are stated in advance. First, with approximately 500 trading days in Track A and twenty assets, the design has limited power against small true differences; wide intervals are the expected presentation, not a defect to be explained away. Second, all inference is conditional on the single evaluation window and universe; no statistical statement in this paper generalizes beyond them, and external validity rests on the design features of Section 3 rather than on the inferential statistics.

5.10 Interpreting Negative, Mixed, or Inconclusive Results

The study’s value does not depend on any strategy outperforming. Pre-committed interpretations: if no LLM-based strategy achieves a corrected positive DSR versus S1, the result will be reported as an absence of evidence for bias-robust value-add in this setting — informative because the setting was designed to remove the artifacts that could manufacture such evidence — and explicitly not as proof that LLM strategies cannot add value elsewhere. If results are mixed across rungs (for example, an informative S5 contrast but an uninformative S6 contrast), the ladder localizes the finding and the paper will report rung-level conclusions only. If results are inconclusive throughout — intervals too wide to distinguish any pair — the paper will report this plainly, present the memorization table (§5.4) and the audit (Section 6) as its primary empirical contributions, and state the sample sizes future work would need. Under no outcome will post-hoc subgroups, alternative metrics, or unplanned configurations be introduced to obtain a reportable difference; any exploratory analysis conducted after unblinding will be labeled as such in a separate, clearly demarcated subsection.

5.11 Contextual Comparison with Published Results

The populated section will close by placing our S6 results alongside the performance reported for FinMem (Yu et al., 2023), FinAgent (Zhang et al., 2024), and the re-evaluation of Li et al. (2025). Because our reference agent, universe, window, and cost model differ from each of these by design, differences in outcomes will be attributed only to named design deltas (agent simplification, cost model, universe rule, contamination condition), and no claim of replication success or failure of any named system will be made.

6. Rationale Audit: Pre-Specified Framework

Registered-report note: as with Section 5, no experiments have been conducted. This section specifies the audit’s constructs, procedures, metrics, and interpretation rules in advance; all tables are placeholders. Design choices, alternatives considered, and remaining limitations are stated explicitly, because the value of an audit depends on the auditor’s methods being open to the same scrutiny as the audited system.

6.1 Constructs and Design Rationale

The audit distinguishes two properties of an explanation that are easily conflated: *plausibility*, whether an explanation convinces a human reader, and *faithfulness*, whether it reflects the actual basis of the model’s decision (Jacovi & Goldberg, 2020). Trading-agent rationales are, by construction, plausible — fluent analyses citing recognizable drivers — but plausibility is precisely what Turpin et al. (2023) show can coexist with systematic misrepresentation of a model’s true decision basis. The audit therefore measures behavioral consistency: whether decisions change under controlled input perturbations in the manner the stated rationales imply. Following Jacovi and Goldberg’s (2020) argument that binary faithfulness is an unrealistic bar, all audit metrics are graded proportions rather than pass/fail judgments. Figure 9 summarizes the audit workflow.

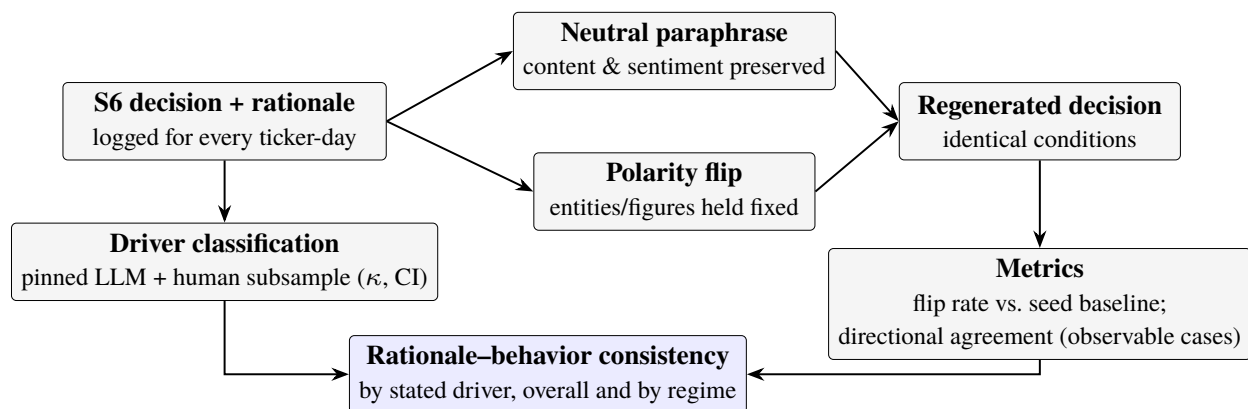


Figure 9: The rationale-audit workflow. Logged rationales are classified by stated driver while perturbed inputs (neutral paraphrase; polarity flip) regenerate decisions; flip-rate and directional-agreement metrics combine with the classification to yield rationale-behavior consistency.

Three alternative designs were considered and rejected. Attribution methods (gradients, attention) attribute token-level influence rather than decision-level drivers — the granularity at which trading rationales speak — and their own faithfulness is contested. Directly querying the model about its reasoning treats the object under suspicion as its own witness (Turpin et al., 2023). Probing internal activations requires white-box instrumentation whose findings are difficult to map onto the trading action space; we note it as a complementary avenue for future work. Input-perturbation testing was chosen because it is model-agnostic, operates at the decision level, and produces directly interpretable quantities. Its principal limitation is stated at the outset: perturbations establish consistency between words and behavior, not ground-truth causation — a rationale could be consistent with behavior yet still not describe the mechanism that produced it. The audit’s conclusions are accordingly framed as consistency findings.

6.2 Rationale Taxonomy and Classification

Every S6 rationale is logged (§4.1) and classified into a taxonomy frozen before experiments: *sentiment-driven*, *price/momentum-driven*, *risk-management-driven*, or *mixed/other*. Classification is performed by the pinned LLM under a fixed prompt, with a random, seeded subsample independently labeled by the author under the budget rule of §6.6. Agreement between machine and human

labels will be reported as a continuous Cohen’s kappa with a bootstrap confidence interval, not as a pass/fail verdict; the pre-specified value $\kappa = 0.70$ serves as an interpretive guideline rather than a statistical truth boundary — if the kappa point estimate falls below it, all analyses use the human labels on the subsample only, with the reduced scope reported, and the kappa value itself is reported in either case. Validation against a single annotator is a limitation; the labeled subsample will be released so that others can re-annotate and re-derive the agreement figure.

To be populated after experiments.

Table 9: Rationale-type prevalence, overall and by regime (placeholder; to be populated after experiments).

Table 9 reports the taxonomy distribution overall and within the regime partitions of §4.6, with observation counts.

6.3 Robustness Test: Neutral Paraphrase

For a pre-specified, seeded random sample of decision days, each news input is paraphrased to preserve content and sentiment while varying surface form; paraphrases are generated by the pinned model under a fixed prompt, and a fixed 10% of audited decisions — selected by seeded sampling stratified across regimes — undergo manual spot checks for content preservation (Appendix D). The decision is then regenerated under identical conditions. The reported metric is the *decision flip rate* under paraphrase. Interpretation rule, fixed in advance: because a paraphrase changes no decision-relevant information, flips indicate sensitivity to surface form; the flip rate is reported with a bootstrap interval and compared against the across-seed flip rate from §4.7, which provides the baseline level of decision instability attributable to sampling nondeterminism alone. A paraphrase flip rate indistinguishable from the seed-only baseline would indicate robustness to wording; an elevated rate would not, and would qualify all rationale-level findings.

6.4 Sensitivity Test: Polarity Flip

For the same sampled days, a second variant reverses the sentiment polarity of the news while holding non-sentiment content as fixed as practicable (entities, figures, and event descriptions retained; evaluative language inverted). The reported metric is the *directional agreement rate*: the proportion of decisions that change in the direction a sentiment-driven reading implies (long toward flat under a negative flip, and conversely). The metric is computed only over decision-days where a response in the implied direction is observable — a flat position cannot respond to a negative flip, nor a long position to a positive one — and the count of excluded, unobservable cases is reported alongside, since silently including them would inflate apparent consistency. The residual confound is acknowledged rather than argued away: inverting evaluative language necessarily alters some content, so polarity flips bound rather than isolate sentiment sensitivity. Construction quality is checked by the same spot-checking procedure as §6.3.

6.5 Rationale–Behavior Consistency

The audit’s central quantity links §6.2 to §6.4: among decisions whose logged rationale is classified sentiment-driven, the proportion that respond to polarity flips; among decisions classified price/momentum-driven, the proportion that do not.

To be populated after experiments.

Table 10: Consistency between stated rationale driver and behavioral response (placeholder; to be populated after experiments).

Table 10 reports these proportions with intervals, overall and by regime — the regime breakdown being descriptive under the same rules as §4.6 and §5.7. Interpretation rules, fixed in advance: high consistency supports treating this agent’s rationales as informative summaries of its behavior *in this setting*; low consistency indicates the rationales should not be relied upon as explanations, which would itself bear on the interpretability claims made for agent architectures (Yu et al., 2023). Neither outcome licenses conclusions about LLM faithfulness in general, about other architectures, or about the mechanism underlying any consistency observed.

6.6 Pre-Specified Scope and Exploratory Extensions

The audit sample size is constrained by inference cost: perturbation tests multiply LLM calls per decision day. The sample size will be determined after the pilot study based on measured inference cost, but the selection rule is fixed in advance: the audit uses the largest regime-stratified, seeded random sample that fits within the pre-specified audit budget while maintaining representation across market conditions. The exact budget and resulting sample size are recorded in Appendix D as protocol parameters — set by the rule and the pilot’s cost measurement, never by examination of results. All analyses in §6.2–§6.5 are pre-specified; any additional analysis suggested by the data will be conducted only after the pre-specified results are fixed and reported in a separate subsection explicitly labeled exploratory, consistent with the frozen-protocol principle of Section 5.

7. Discussion: Pre-Committed Interpretive Framework

Registered-report note: this section fixes the interpretive frame before results exist. Each subsection states, conditionally, what the principal possible outcomes would and would not imply. When results are available, this section will be rewritten in declarative form, bound by the commitments below; any interpretation not anticipated here will be identified as post hoc. Expected implications — consequences of the design regardless of outcome — are distinguished throughout from empirical conclusions, which exist only after experiments.

7.1 Interpreting the RQ1 Outcome

Three principal outcomes are possible, and the appropriate reading of each is fixed in advance. If no LLM-based strategy achieves a corrected positive DSR against the passive benchmark under any contamination condition, the result would align with the skeptical re-evaluation of Li et al. (2025) and extend it: not only do reported advantages fade under broader evaluation, they fail to appear even in a setting purpose-built to detect them if they exist — while remaining, per §5.10, an absence of evidence confined to this universe, window, and reference agent. If an LLM-based strategy’s advantage survives the full set of controls — anonymization, cost stress, multiple-testing correction, and directional confirmation in Track B — the result would constitute the first bias-controlled evidence of LLM trading value of which we are aware, and its credibility would rest precisely on the severity of the controls it survived; the ladder position of the surviving rung (scorer versus agent) would localize the source. If an advantage appears under original inputs but disappears under anonymization, the natural interpretation is that apparent skill reflects memorization or entity knowledge rather than signal extraction — the outcome with the most direct implications for how prior backtests should be read, and the one the three-condition design is constructed to isolate at the strategy level, extending the scoring-level isolation of Glasserman and Lin (2023). In all three cases, what would *not* be justified is identical: no generalization to other universes, frequencies, model families, or live trading.

7.2 Methodological Implications (Design-Level, Outcome-Independent)

One implication attaches to the design rather than to any result. If the memorization-quantification protocol of §3.4 proves executable at reasonable cost — an empirical question the pilot answers — then historical backtests of LLM strategies that omit such controls are, at minimum, incomplete as evidence, since a low-cost procedure existed to check whether their performance survives anonymization. This is an argument about evidentiary standards, not about any particular system’s honesty, and it echoes the trajectory of the earlier backtest-validity literature, in which corrections such as the deflated Sharpe ratio migrated from methodological proposals to reviewer expectations (Bailey & López de Prado, 2014). Whether our specific implementation merits adoption depends on the Track A–Track B agreement reported under §5.6, and we commit to reporting that agreement whether it supports or undermines the protocol.

7.3 Interpreting the RQ2 Outcome

If rationale–behavior consistency is high, the defensible conclusion is that this agent’s stated rationales are usable summaries of its behavior in this setting — supporting, in a limited way, the interpretability benefit claimed for agent architectures (Yu et al., 2023), while establishing nothing about mechanism (§6.1). If consistency is low, the implication is more consequential: rationales that read as explanations but do not track behavior are a liability rather than a feature, inviting misplaced trust from users and reviewers alike, and consistent with the unfaithfulness phenomena documented outside finance (Turpin et al., 2023). An intermediate or regime-varying pattern would be reported descriptively and framed as hypothesis-generating. Under every outcome, the audit speaks to one reference agent and one model; claims about agent interpretability in general would require replication across architectures, which we do not perform.

7.4 Theoretical Reading

The regime analysis, being descriptive, supports interpretation but not testing. Directional patterns will be read against two frames fixed in advance: under the efficient markets hypothesis, no strategy run should retain value net of costs in so heavily arbitrated a universe (Fama, 1970); under the adaptive markets hypothesis, any observed value should be episodic and concentrated in stressed conditions (Lo, 2004), consistent with the recession concentration documented by García (2013) and the bear-market asymmetry reported by Li et al. (2025). Whichever pattern the data show, the reading remains interpretive: this study’s design can render one frame more or less congenial, but a two-year window on twenty stocks adjudicates between market-efficiency theories no more than any single backtest can, and the paper will say so.

8. Limitations

The limitations below are boundaries on the conclusions this study can support, not defects discovered after the fact; each was accepted as a design trade-off and is paired with the section that manages it.

Data. The news corpus samples four financial websites (Dong et al., 2024) and therefore represents one slice of the information environment; strategies are evaluated relative to this information set, not to all news that existed (§3.1). Point-in-time correctness depends on source timestamp accuracy, which we test but cannot fully guarantee; where granularity is coarse, the conservative execution convention of §3.2 protects integrity at the cost of signal freshness. Track B relies on free news APIs whose firm-level coverage and precision are verification items, and its six-month span limits it to directional validation. Residual contamination risk remains even under the three-condition protocol: anonymization can leave stylistic or contextual cues that identify entities, and no procedure can prove that memorized knowledge exerted no influence — which is precisely why Track A and Track B are designed to check one another (§3.4, §5.6).

Modeling. All LLM results are conditional on a single pinned open-weight model and on the specific frozen prompts of Appendix A; we do not explore prompt variations or model families, so conclusions attach to this instrument, not to “LLMs” as a class (§4.7). Stochastic decoding is bounded by the three-seed dispersion analysis rather than eliminated. Most fundamentally, the rationale audit measures consistency between stated reasons and behavior; it cannot establish that a rationale describes the causal mechanism of a decision (§6.1), and no claim in this paper depends on such a reading.

Evaluation. A backtest is a simulation: it omits market impact, execution latency, capacity constraints, and the feedback effects of actual participation, and no result here guarantees or predicts live performance. The primary window spans two years in a single market and its most liquid segment — a deliberately conservative testbed (§3.1) that limits generalization in the opposite direction as well: a null here does not preclude value in less efficient segments. Statistical power against small effects is limited, intervals will be wide, and the regime analysis is descriptive by design (§4.5, §4.6); financial inference from a single window is inherently uncertain, which the pre-committed interpretation rules of §5.10 and §7.1 are constructed to respect rather than obscure.

Audit. Human validation of rationale classification rests on a single annotator, with a released subsample as the remedy (§6.2); perturbation quality is spot-checked on a fixed 10% fraction rather

than exhaustively; polarity flips necessarily alter some content, so sentiment sensitivity is bounded rather than isolated (§6.4); and the audited sample is budget-constrained (§6.6). Audit findings therefore characterize one agent under one model in one setting, and their proper interpretive weight is fixed in §7.3.

9. Future Work

The extensions below follow from the limitations; we distinguish direct, realistic continuations from more speculative directions. Realistic near-term extensions: repeating the protocol on mid- and small-capitalization universes, which are plausibly less efficient than the deliberately conservative mega-cap testbed used here; extending the horizon toward multi-decade evaluation in the manner of Li et al. (2025); running the frozen protocol across several model families to test instrument dependence; full-fidelity replication of named agent systems, which our simplified reference agent deliberately does not attempt; multi-annotator validation of the rationale taxonomy; and a longer forward (paper-trading) window, which strengthens Track B’s role at the cost of calendar time. More speculative directions, contingent on tooling and access: white-box causal analysis of agent decisions via activation-level intervention, which could convert the audit’s consistency findings into mechanism findings; extension to other asset classes such as cryptocurrencies or fixed income, where information environments differ qualitatively; and study of market-level effects if LLM-driven strategies become widespread — a question this single-agent design cannot touch. Any real-world deployment consideration would additionally require the market-impact, capacity, and latency modeling that backtesting omits, and none of this study’s outputs should be read as deployment guidance.

10. Conclusion (Results-Pending Framework)

Registered-report note: to be rewritten in declarative form once experiments conclude; the commitments below bound that rewriting.

This study addresses a contradiction at the center of the LLM-trading literature: developer-reported backtests that beat the market, and independent re-evaluation under which those advantages disappear. Its response is methodological — a pre-specified, bias-controlled evaluation framework comprising a six-strategy ladder on identical data, a three-condition memorization-quantification protocol, an out-of-cutoff validation track, corrected performance statistics, and a rationale audit that tests whether agents’ explanations track their behavior. The framework is designed to determine where, if anywhere, along the path from price signal to autonomous agent value is added; how much of apparent historical performance memorization explains; and whether stated rationales can be trusted as summaries of agent behavior. The paper claims no strategy outperforms any other, and no hypothesis is validated by anything reported here; every such conclusion depends on the experiments specified in Sections 3–6 and will be stated, when available, under the interpretation rules fixed in Sections 5 and 7. If the eventual results survive the framework’s controls, their significance will lie in being the first such evidence produced under conditions designed to exclude the known artifacts; if they do not, the significance will lie in showing that a purpose-built, low-cost protocol suffices to separate evaluation artifact from

genuine capability — and, in either case, in the released code, cached model outputs, and frozen protocol that allow others to check both.

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Appendix A: Frozen Prompts (version A-1.0)

All five roles use the pinned model with `enable_thinking=false` and the generation parameters of Appendix D. Structured outputs are schema-validated; a malformed response is retried once, then logged as an abstention (§4.7).

A.1 S5 sentiment scorer. “You are a financial news analyst. Given the news items below about one company, output JSON {”score”: s} where $s \in [-\$1, 1]$ reflects the net sentiment of the news for the company’s near-term stock performance. Consider only the provided text. Do not use any other knowledge about the company.”

A.2 S6 trading agent. “You are a trading assistant managing a single-stock position. Inputs: recent news summaries (trailing 3 trading days), daily closing prices (trailing 60 trading days), current position (long/flat). Decide the position for the next session. Output JSON {”action”: ”long”|”flat”, ”rationale”: one paragraph explaining the primary driver of your decision}. Base your decision only on the provided inputs.”

A.3 Rationale classifier. “Classify the primary driver stated in this trading rationale into exactly one of: sentiment, price_momentum, risk_management, mixed_other. Output JSON {”label”: ...}. Judge only what the rationale states, not whether it is correct.”

A.4 Paraphrase generator. “Rewrite the following news text preserving all factual content, named entities, figures, and sentiment. Change only wording and sentence structure. Output the rewritten text only.”

A.5 Polarity-flip generator. “Rewrite the following news text keeping all named entities, figures, and event descriptions unchanged, but invert the evaluative language so the overall sentiment is reversed. Output the rewritten text only.”

Appendix B: Hyperparameters and Calibration (summary)

S2: momentum lookback 252 trading days, skip 21, monthly signal refresh. S3/S5: threshold = Sharpe-maximizing percentile of the pooled calibration-window (2019-01-02 to 2021-12-31) score distribution from {30, 40, 50, 60, 70}; ties to the lowest percentile. S4: LightGBM (seed 42), features {lagged returns 1/5/21 d; MA ratios 10/50 and 50/200; 21-day realized volatility; 21-day volume z-score; sentiment daily and trailing-3-day mean}; search space $\text{num_leaves} \{15, 31, 63\} \times \text{learning_rate} \{0.01, 0.05, 0.1\} \times \text{n_estimators} \{100, 300, 500\} \times \text{min_child_samples} \{20, 50\} \times \text{feature_fraction} \{0.8, 1.0\}$; 5-fold expanding-window CV, log-loss criterion; decision rule long iff $P(\text{up}) > 0.5$. Full machine-readable values: repository `configs/`.

Appendix C: Per-Ticker Results

All strategies and conditions, reported in full (to be populated after experiments; no selective emphasis).

Appendix D: Frozen Experimental Protocol

D.1 Model instrument. Qwen3-14B (instruction-tuned), Hugging Face repository Qwen/Qwen3-14B (Apache-2.0; 14.8B parameters; 32,768-token context); weights revision hash and download date recorded at download [mechanical, pre-pilot]. Inference: vLLM (version recorded at download), single GPU, bf16, no quantization (llama.cpp Q8_0 fallback only via amendment). Generation: temperature 0.7, top-p 0.8 (Amendment A-001), role-specific max_tokens (S6 512; S5 scorer 64; classifier 64; perturbation generators 1024), fixed stop sequences, seeds {11, 42, 2026}, enable_thinking=false. Malformed-output policy: retry once, then abstain (hold position; abstentions counted). No model substitution after results are seen; runs with any other model are labeled exploratory.

D.2 Data snapshots. FNSPID repository commit hash and file checksums recorded at download; universe = twenty largest S&P 500 constituents by market capitalization as of 2021-12-31 (tickers listed at pilot, frozen thereafter); Track A window 2022-01-03 to 2023-12-29; Track B window 2026-01-02 to 2026-06-30; calibration window 2019-01-02 to 2021-12-31; prices via yfinance (version and adjustment settings recorded) with an independent second source for cross-checks [PILOT→PARAM]; risk-free rate FRED series DTB3; S&P 500 total return ^SP500TR (Yahoo Finance); VIX daily closes (Cboe/Yahoo, source recorded); Fama–French daily factor file version recorded at download; Track B news query parameters (Finnhub company-news; GDELT filters) [PILOT→PARAM].

D.3 Strategy parameters. As in Appendix B; S6 price-history window 60 trading days; news window 3 trading days (matching the M1 criterion); action space {long, flat}.

D.4 Evaluation grid and statistics. Costs 5/10/25 bps per side; contamination conditions original / anonymized / anonymized+shuffled dates; DSR trial count = all strategy–cost–condition combinations executed; stationary bootstrap B = 10,000, expected block length 21 trading days (sensitivity at 5 and 63 reported), 95% intervals, seed 2026; regime rules as in §4.6.

D.5 Audit parameters. Total inference budget USD 200; audit share 30%; audit sample = largest regime-stratified seeded sample within budget [PILOT→PARAM]; human-validation subsample min(200, 20% of audited rationales); $\kappa = 0.70$ interpretive guideline with continuous reporting; spot-check fraction 10%; taxonomy {sentiment, price_momentum, risk_management, mixed_other}.

D.6 Pilot. Two tickers (seeded random draw), three-month sub-window; verification targets per §3.5 including the frozen news-density criterion ($M1 \geq 0.90$ on the trailing-3-day window; $M2 \geq 0.95$; >5/20 failures → amendment); pilot outcomes may amend the protocol only through the Amendment Log, before full-scale runs; pilot trading performance is never reported as evidence.

D.7 Recorded implementation conventions. Entry decisions are dated the last trading day before the evaluation window (positions effective at the window’s first open). Ticker-days without news map to flat (S3/S5); incomplete feature rows map to flat and missing sentiment features to neutral zero (S4); zero-variance volume windows yield $z = 0$. S4 labels are the direction of the $\text{open}(t+1) \rightarrow \text{open}(t+2)$ return (D-S4-LABEL). Downside deviation is the annualized target-0 Sortino denominator (resolves D-METRICS-1). Timestamp-granularity execution convention per §3.2 [PILOT→PARAM]. Calibration and grid-search ties resolve to the lowest percentile / first combination in deterministic iteration order.

D.8 Release artifacts. Code repository (harness, strategies, analysis), version-controlled run

configurations with logged protocol-lock hash, cached LLM calls (prompts and outputs), labeled rationale subsample, and data-reconstruction scripts.

Status at manuscript freeze: all rules and thresholds above are frozen. Items marked [PILOT→PARAM] and download-time recordings (model revision, dataset commits, library versions, universe ticker list) receive values before the first full-scale experiment; no experiment runs while any such field is empty, enforced mechanically by the repository’s protocol-lock check.

Appendix E: Anonymization Procedure

For each universe firm, a curated alias dictionary (company names, tickers, brands and products, executive names, headquarters city) is compiled from public profiles and frozen before the pilot. Replacement is longest-alias-first, case-insensitive, and word-boundary-anchored; URLs, bylines, and publication names are stripped first. Placeholders are consistent within each decision day and re-randomized across days (seed 42), so cross-day identity cannot be inferred. A spaCy named-entity-recognition pass (version recorded) masks residual ORG/PERSON/GPE entities the dictionaries miss. Date shuffling permutes article timestamps within ticker across the Track A window (seed 42), preserving the article set exactly. Validation: on the 10% spot-check sample, the annotator attempts firm identification from anonymized text, and the identification rate is reported as a protocol-quality metric (not tuned against).

Amendment Log

A-001 (2026-07-09) — instrument configuration correction. Change: `generation.temperature` 0.0 → 0.7; `generation.top_p` 1.0 → 0.8. Reason: model validity — the Qwen3 documentation advises against greedy decoding, and the documented non-thinking-mode inference configuration provides a more valid evaluation of the instrument. Reproducibility remains enforced through fixed seeds, full prompt/output caching, revision pinning, and experiment logging. Affected analyses: all LLM-based runs (none executed at amendment time). Outcome-triggered: no.

Ethics and use note: this study uses simulated backtesting only; no live trading is conducted, and nothing in this paper constitutes investment advice. No human subjects are involved beyond the author’s annotation of model outputs.